System and Devices (SYS 3)

Lecture 16: Block Replacement and Write Strategy

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COM00018I

Four cache memory design choices

- Where can a block be placed in the cache?
  - Block Placement
- How is a block found if it is in the upper level?
  - Block Identification
- Which block should be replaced on a miss?
  - Block Replacement
- What happens on a write?
  - Write Strategy

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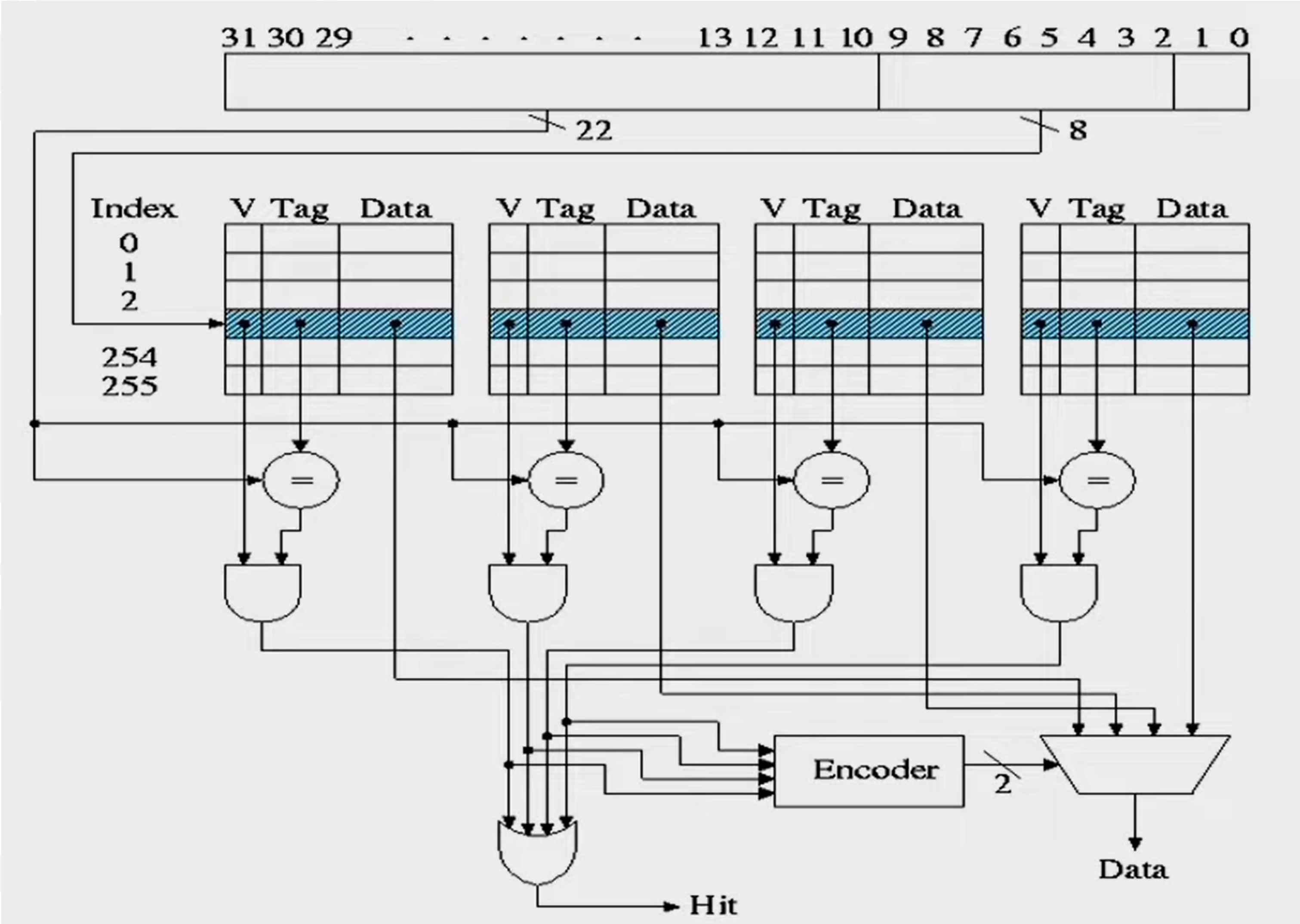
Block Replacement

- Cache has finite size. What do we do when it is full?
- Direct Mapped is easy.
- Which block to be replaced for a set associative cache?



Block Replacement Algorithms

- Random
- First In First Out (FIFO)
- Last In First Out (LIFO)
- Least Recently Used (LRU)
- Pseudo-LRU (PLRU)
- Not Recently Used (NRU)
- Lease Frequently Used (LFU)
- Re-Reference Interval Prediction (RRIP)
- Optimal

Random & FIFO Replacement Policy

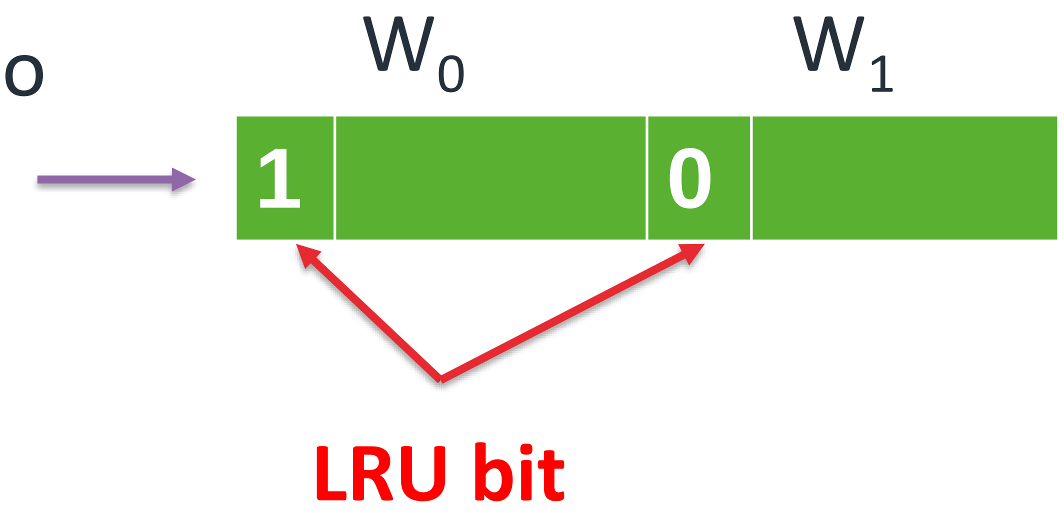
- **Random policy** needs a pseudo-random number generator
- Makes no attempt to take advantage of any temporal or spatial localities
- **First-in, Fist-out (FIFO) policy** evict the block that has been in the cache the longest
- It requires a queue Q to store references
- Blocks are enqueued in Q, dequeue operation on Q to determine which block to evict

Least-Recently Used

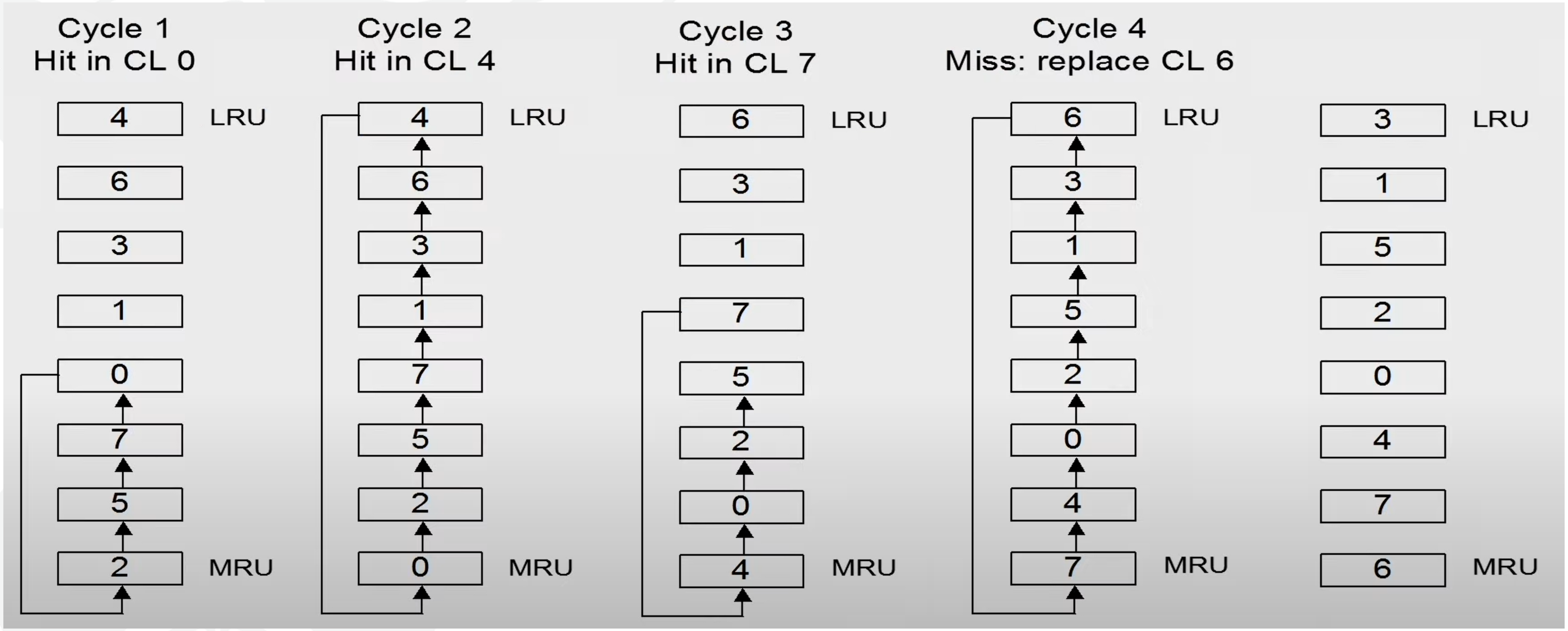
- For associativity = 2, LRU is equivalent to NMRU
  - Single bit per line indicates LRU/MRU
  - Set/clear on each access

For a>2, LRU is difficult/expensive

- Record Timestamps? How many bits?
- Must find min timestamp on each eviction
- Sorted list? Re-sort on every access?
- Shift register implementation



LRU Implementation



Optimal Replacement Policy

- Evict block with longest reuse distance
  - i.e., next reference to block is farthest in future
  - Requires knowledge of the future!

Can't build it, but can model it with trace

Useful, since it reveals opportunity

Optimal better than LRU

- (X,A,B,C,D,X): LRU 4-way SA cache, 2nd X will miss

LRU:

C	⇨	C
B		B
A		X
X		D

Optimal Replacement Policy

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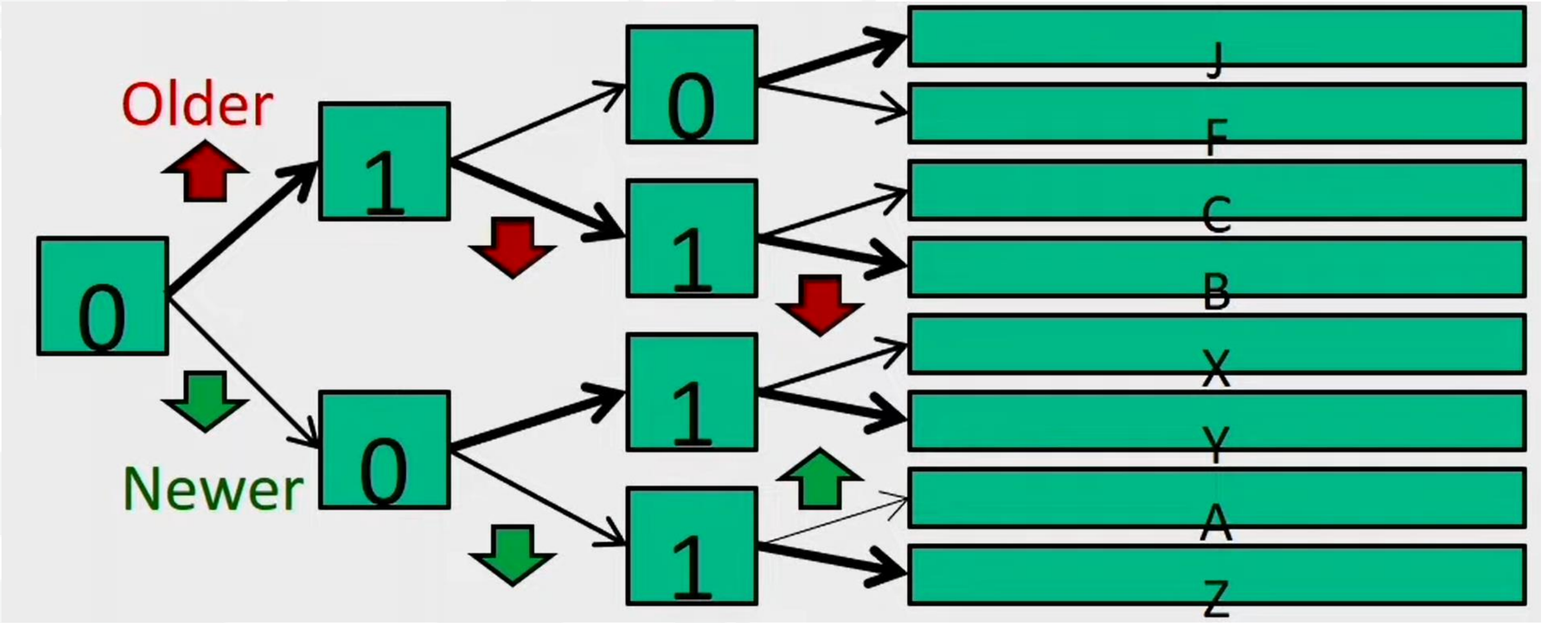
- (X,A,B,C,D,X): LRU 4-way SA cache, 2nd X will miss

Optimal:

C		C
B	⇨	B
A		D
X		X

Practical Pseudo-LRU



- Rather than true LRU, use binary tree
- Each node records which half is older/newer
- Update nodes on each reference
- Follow older pointers to find LRU victim

Not Recently Used (NRU)

- Keep NRU state in 1 bit/block

    - Bit is reset to 0 when installed / re referenced

    - Bit is set to 1 when it is not referenced and other block in the same set is referenced

    - Evictions favor NRU=1 blocks

    - If all blocks are NRU=0 / 1 then pick by random

- Provides some scan and thrash resistance
- Randomizing evictions rather than strict LRU order

Re-reference Interval Prediction

- RRIP
- Extends NRU to multiple bits
  - Start in the middle
  - Promote on hit
  - Demote over time
- Can predict near-immediate, intermediate, and distant re-reference

Least Frequently Used

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- Counter per block, incremented on reference
- Evictions choose lowest count
- Logic not trivial (a^2 comparison/sort)
- Storage overhead
  - 1 bit per block: same as NRU
  - How many bits are helpful?

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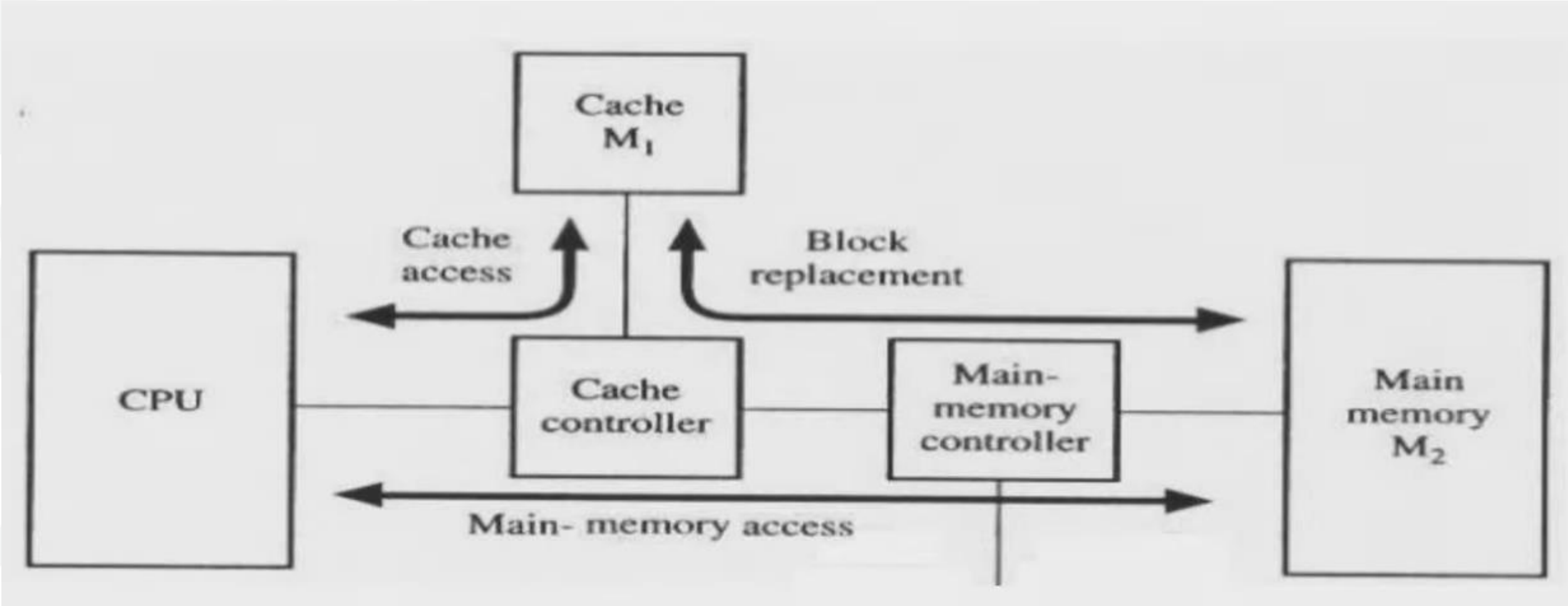
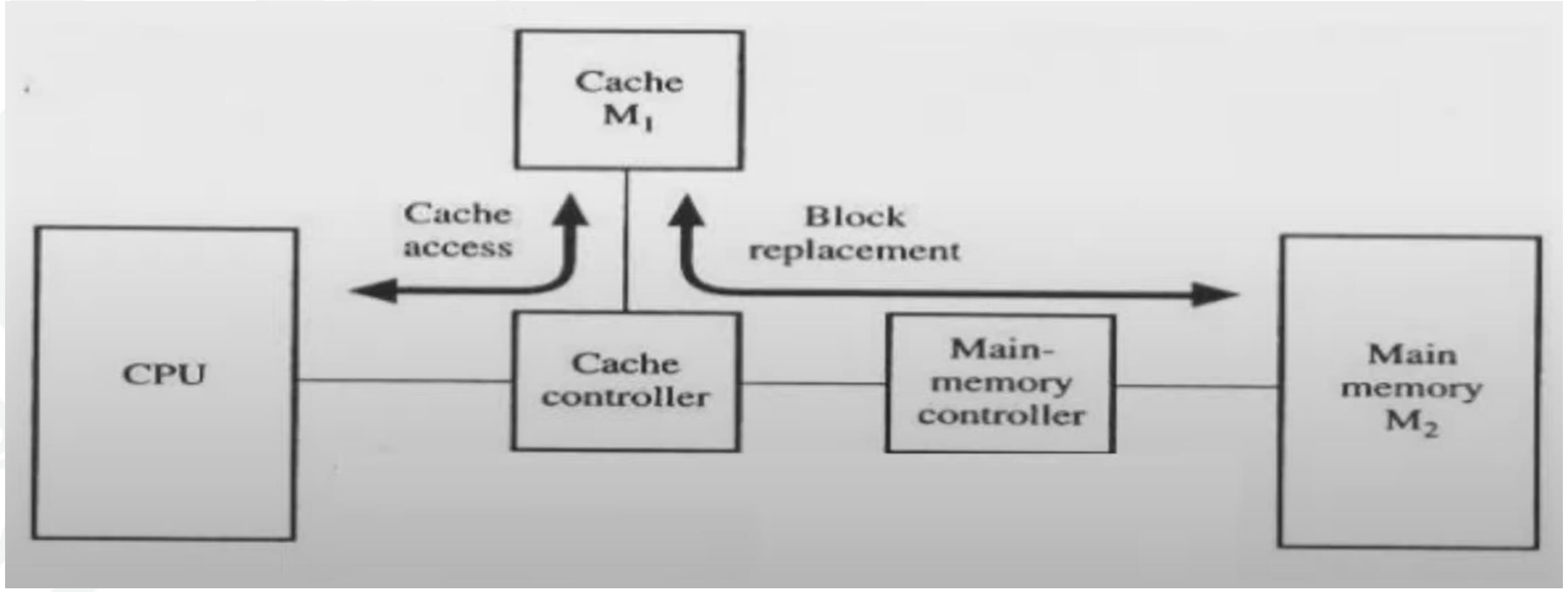
Look-aside vs Look through caches

- **Look-aside cache**: Request from processor goes to cache and main memory in parallel
- Cache and main memory both see the but cycle
- On cache hit ➡ processor loaded from cache, bus cycle terminates; On cache miss ➡ processor & cache loaded from memory in parallel



Look-aside vs Look through caches

- **Look-through cache**: Cache checked first when processor requests data from memory
- On cache hit ➡ data load from cache; On cache miss ➡ cache loaded from memory, then processor loaded from cache

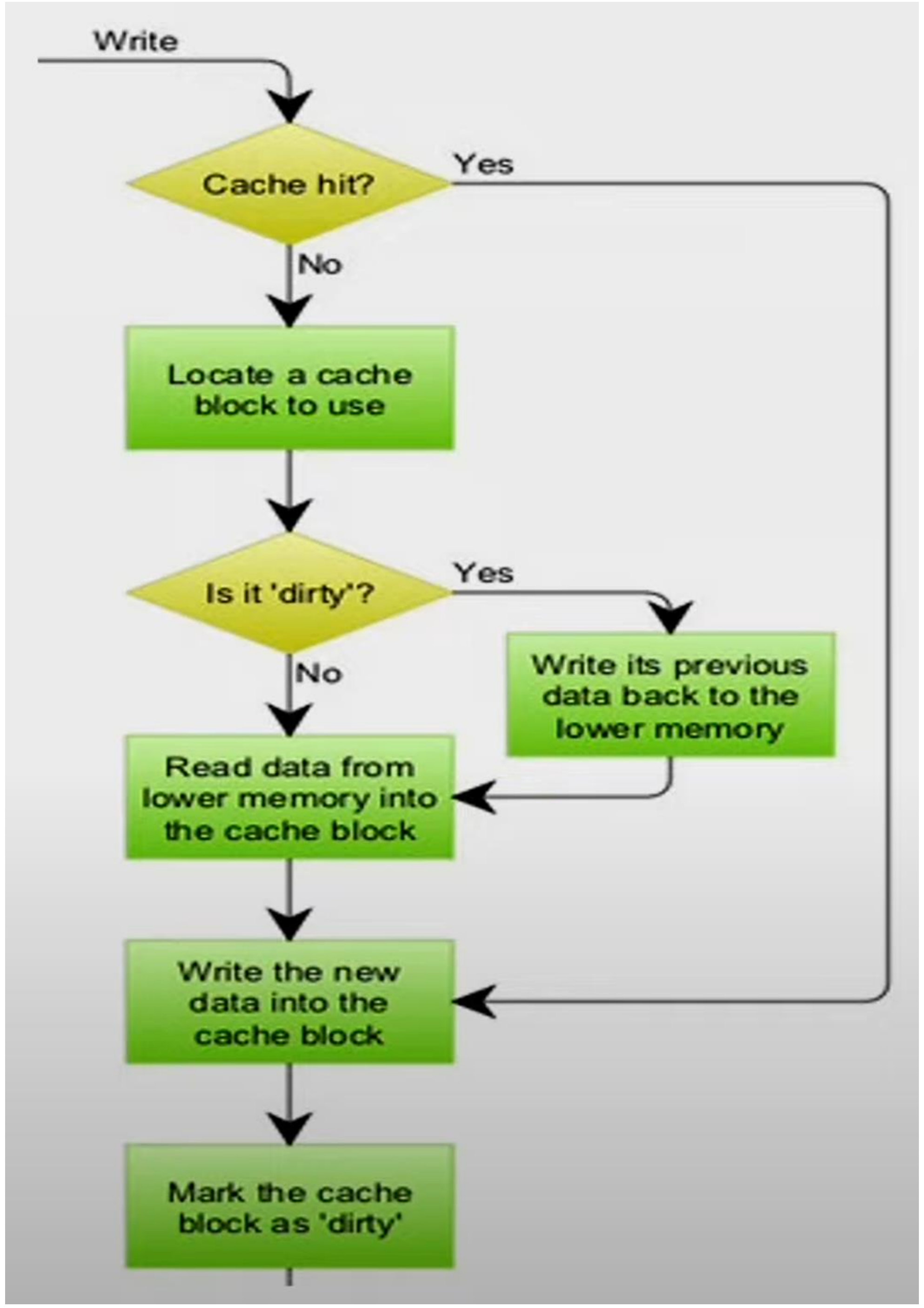


Write Strategy

- Write Hits → Write through vs Write back
- Write Miss → Write allocate vs No-Write allocate
- Write through: The information is written to both the block in the cache and to the main memory
- Read misses do not need to write back evicted line contents
- Write back: The information is written only to the block in the cache. The modified cache block is written to main memory only when it is replaced.
- Have to maintain whether block clean or dirty. No extra work on repeated writes; only the latest value on eviction gets updated in main memory

Write Strategy

- **Write allocate**: The block is loaded into cache on a write miss.
- Used along with write back caches



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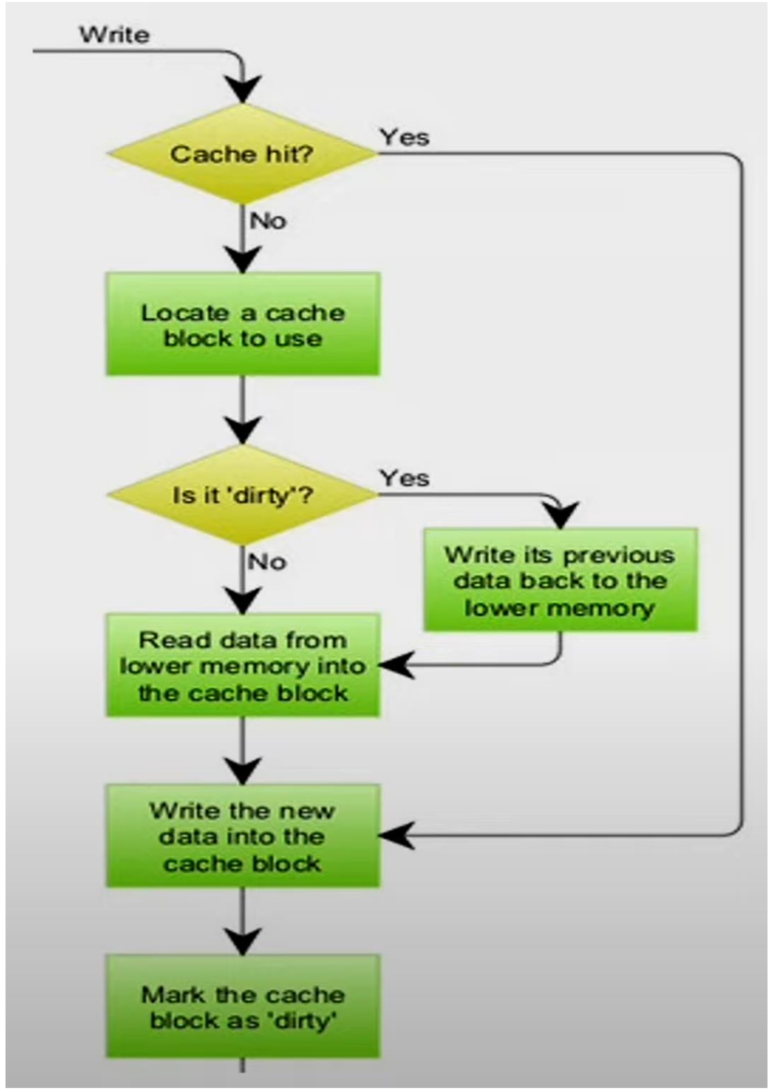
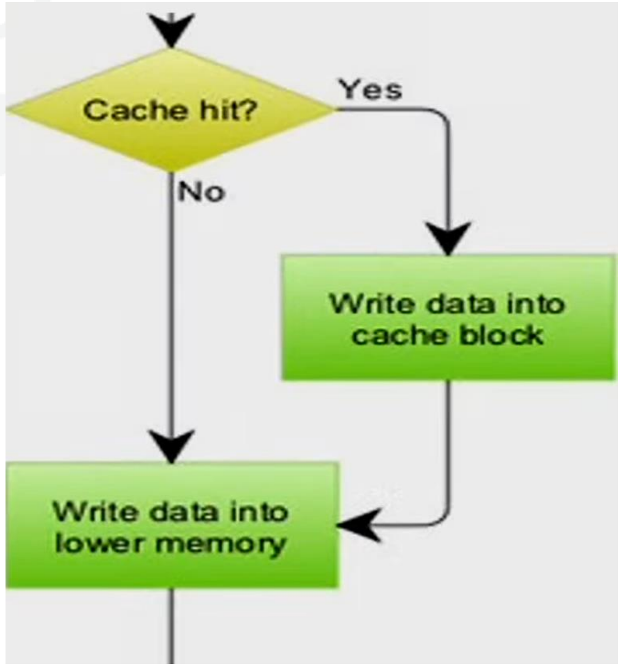
Write Strategy

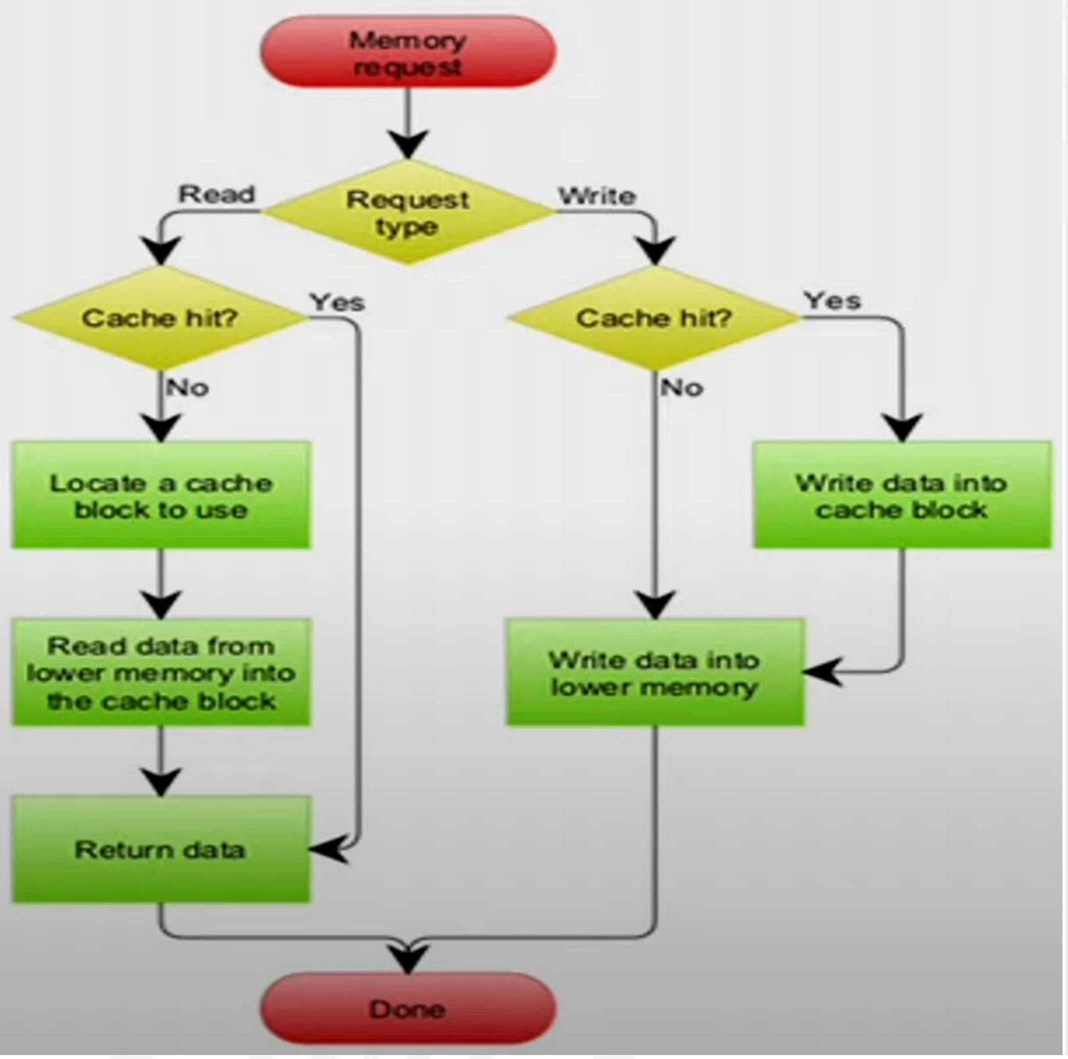
- **Write allocate**: The block is loaded into cache on a write miss.
- Used along with write back caches
- **No-Write allocate**: The block is modified in the main memory but not in cache
- Used along with write through caches



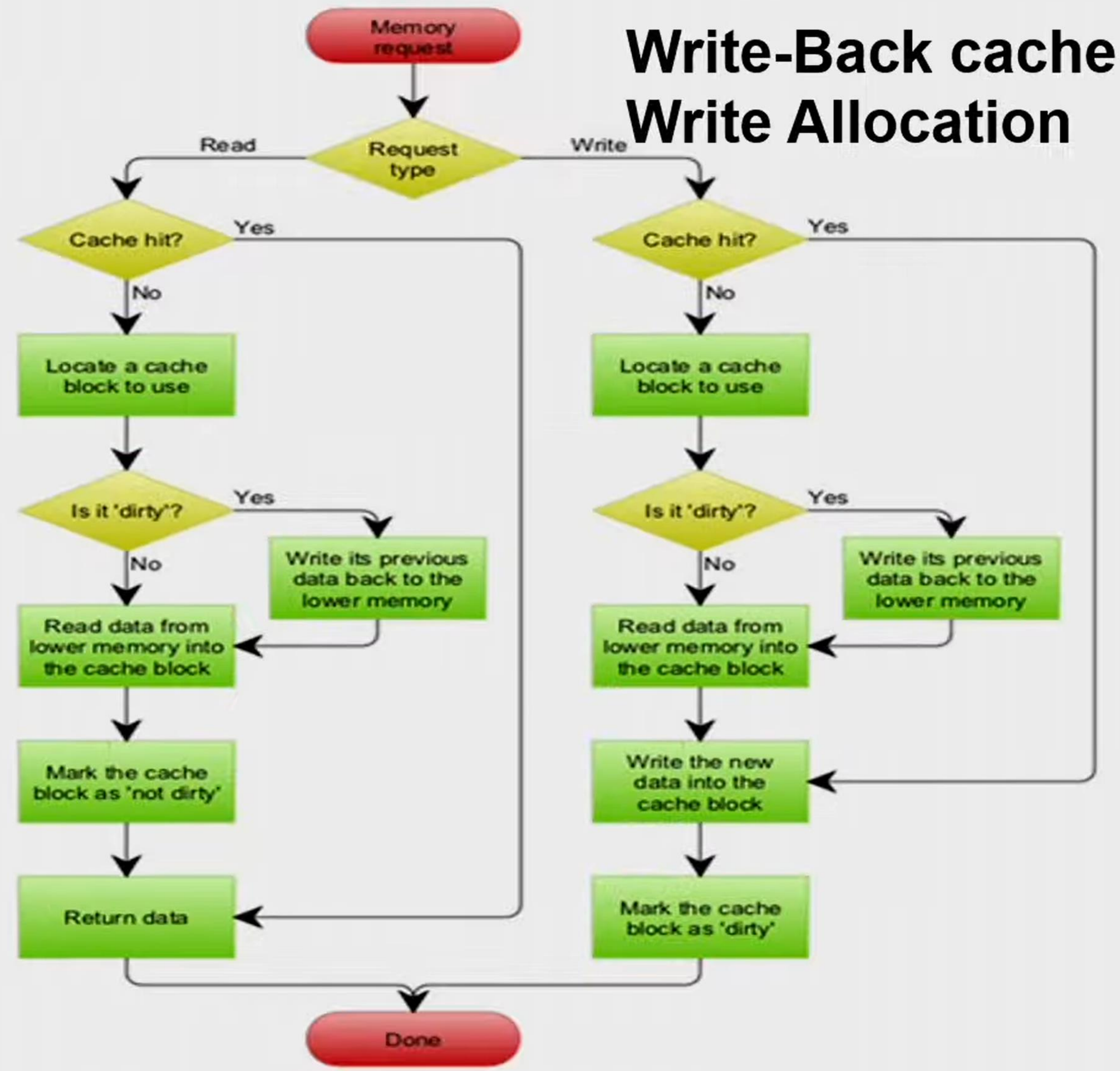


Write Strategy

Write-Through cache with No-Write Allocation



Write-Back cache with Write Allocation



Types of Cache Misses

- **Compulsory**
  - Very first access to a block
  - Will occur even in an infinite cache
- **Capacity**
  - If cache cannot contain all the blocks needed
  - Misses in fully associative cache (due to the capacity)
- **Conflict**
  - If too many blocks map to the same set
  - Occurs in associative or direct mapped cache

Suggested Readings

Computer Architecture: A Quantitative Approach (5th Edition) by John Hennessy and David A Patterson:

- pp. 72 – 78: Section 2.1
- pp. B6 – B12: Appendix B Review of Memory Hierarchy

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